

Assignment 2

Part 1: Zipf's Law and Corpus Analysis

Corpus used: Brown Corpus (NLTK) — 1,000,000 tokens, 46,207 types.

The provided English dataset (~152K tokens) is below the 1 million token requirement, so the Brown corpus (via NLTK) is used.

Q1: Top-10 Words and Frequencies

Rank	Word	Frequency
1	the	62,169
2	,	49,603
3	.	40,669
4	of	33,360
5	and	24,709
6	to	22,854
7	a	19,854
8	in	19,154
9	is	9,728
10	that	9,196

Q2: Zipf's Law — Log-Log Plot and Linear Regression

A linear regression was fitted to $\log_{10}(\text{rank})$ vs $\log_{10}(\text{frequency})$.

Slope = -1.2994, Intercept = 5.9497. Zipf's Law predicts a slope of approximately -1.

The corpus deviates from this prediction (observed slope $\neq$ -1). The slope being steeper than -1 indicates that high-frequency words occur even more dominantly than a pure Zipf distribution would predict, which is typical for English text where function words and punctuation create a heavy head.

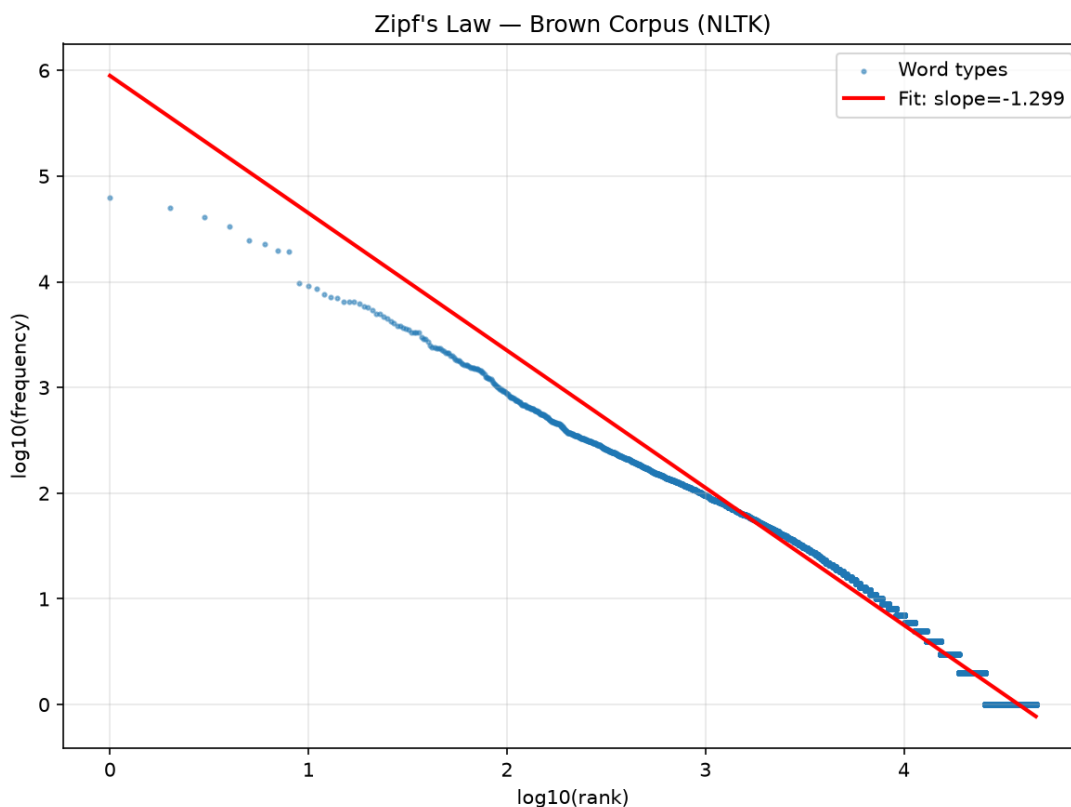


Figure 1: Zipf's Law log-log plot with regression line

Q3: Top-20 and Bottom-20 Word Types

Top-20 word types: the, ,, ., of, and, to, a, in, is, that, for, was, it, he, as, ``', " , with, be, on

Bottom-20 word types: implant, senselessly, hoarsely, child-face, dabbed, toe-tips, teratologies, nicest, burned-out, ground-truck, threshed, bawled, perplexity, cow-man, cow-people, resembling, kindness, geysers, commingled, b'dikkat's

Comment on grammatical categories:

The top-20 words are entirely closed-class (20/20): determiners (the, a), prepositions (of, in, to), conjunctions (and), pronouns (he, it), and punctuation. The bottom-20 words are predominantly open-class (13/20): rare nouns (implant, geysers), verbs (dabbed, threshed), and adjectives (nicest, kindness).

This connects to the type/token distinction: a few closed-class types account for a large share of all tokens (high token count, low type diversity at the head), while open-class words form the long tail of low-frequency types. Many content-word types appear only once (hapax legomena), contributing one token each.

Q4: Type-Token Ratio (TTR) vs Corpus Size

Corpus Size (tokens)	TTR
10,000	0.2507

50,000	0.1640
100,000	0.1308
500,000	0.0662
1,000,000	0.0462

Discussion: TTR decreases monotonically as corpus size grows (0.251 at 10K $\rightarrow$ 0.046 at 1M). Each new token block introduces previously unseen word types, but vocabulary growth slows (Heaps' Law) while token count increases linearly. Therefore $|V|/N$ decreases. Raw TTR is not suitable for comparing texts of different lengths.

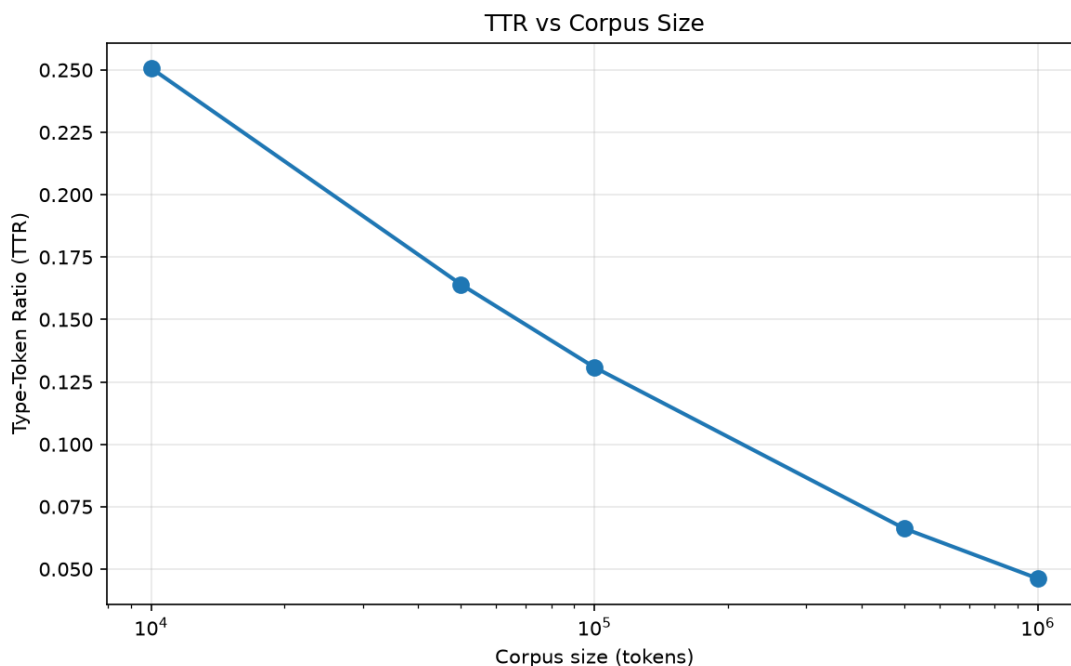


Figure 2: TTR vs corpus size

Q5: Standardized TTR — MATTR

Corpus Size	Raw TTR	MATTR
10,000	0.2507	0.5015
50,000	0.1640	0.5113
100,000	0.1308	0.5127
500,000	0.0662	0.4997
1,000,000	0.0462	0.4834

Method implemented: MATTR (window=500). MATTR computes TTR over a sliding window of fixed size and averages the results. Unlike raw TTR, MATTR remains stable (~ 0.48 – 0.51) across corpus sizes because each measurement uses the same window length. This makes MATTR a more reliable metric for cross-corpus comparison.

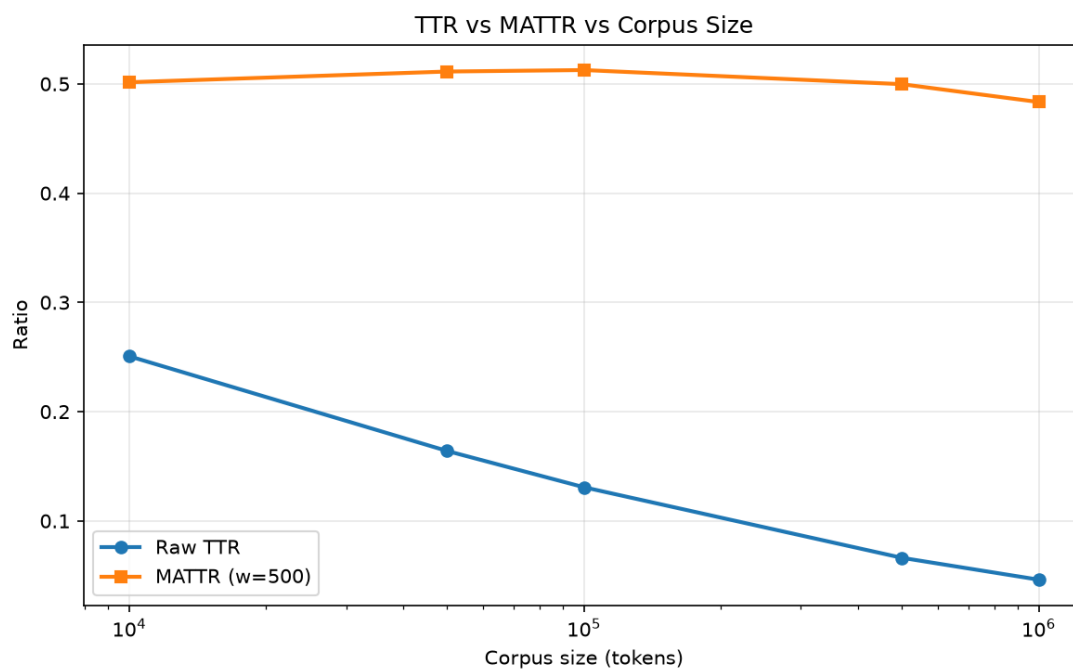


Figure 3: Raw TTR vs MATTR vs corpus size

Part 2: Language Models

Corpus: Provided English dataset (train+dev+test, 80/20 split)

Training sentences: 9,600 | Test sentences: 2,400

Split: 80% train / 20% test. Sentence boundary tokens <s> and </s> added to each sentence.

Q6: MLE Unigram and Bigram Probabilities

Unigram probabilities $P(w)$ and bigram probabilities $P(w_i|w_{i-1})$ were estimated from the training set using Maximum Likelihood Estimation (MLE):

$$P(w) = \text{count}(w) / N$$

$$P(w_i|w_{i-1}) = \text{count}(w_{i-1}, w_i) / \text{count}(w_{i-1})$$

Vocabulary size (including <s>, </s>, <unk>): 5,335

Distinct bigram types observed: 51,830

Q7: Add-1 (Laplace) Smoothing

Why is raw MLE insufficient for unseen bigrams?

Raw MLE assigns $P(w_i|w_{i-1}) = 0$ for any bigram not observed in training. On the test set, a single unseen bigram makes the entire sentence probability zero, rendering perplexity undefined ($\log 0 = -\infty$).

What problem does Laplace smoothing solve?

Add-1 smoothing assigns non-zero probability to unseen events:

$$P(w_i|w_{i-1}) = (\text{count}(w_{i-1}, w_i) + 1) / (\text{count}(w_{i-1}) + |V|)$$

For seen bigrams, MLE estimates are used; Laplace is applied only when count = 0. This ensures all test sentences receive a valid, non-zero probability.

Q8: Perplexity on Held-Out Test Set

Model	Perplexity
Unigram (MLE)	346.50
Bigram (Laplace)	165.88

The bigram model conditions on the previous word, capturing local syntactic and lexical dependencies (e.g., 'the' is often followed by nouns). This produces sharper, more accurate probability distributions than the unigram model, which treats each word independently. Lower perplexity means the model is less 'surprised' by the test data.

Q9: Five Random Sentences (Bigram Model, Ancestral Sampling)

- <s> child waters and took fear wherever they brahmins and watch exist </s>
- <s> so accused muscles </s>

3. <s> specify resource brass communicate superlibrarian clothing top of shlokas nights executes completes one who are eating wheel toward communication auspicious able juda the yogi is hand importance of <unk> of </s>
4. <s> to idols we shall come unto lizard tail follows administrators feather </s>
5. <s> festus gopal journey generation </s>

Q10: Trigram Model Comparison

Model	Perplexity
Unigram (MLE)	346.50
Bigram (Laplace)	165.88
Trigram (Laplace)	562.75

Does the trigram model have higher perplexity than the bigram model? Yes.

Discussion:

1. Sparsity: Trigram contexts (w_{i-2}, w_{i-1}, w_i) are far rarer than bigram contexts. Most trigrams in the test set were never seen in training.
2. Smoothing: Laplace smoothing spreads probability mass over $|V|$ outcomes for each context, which is more harmful for trigrams because the denominator is large relative to observed counts.
3. Longer history: Conditioning on two previous words increases model capacity but also data sparsity. With a limited corpus (~12K sentences), the trigram model overfits observed patterns and generalizes poorly to held-out data.

Part 3: HMM POS Tagger with Viterbi

Corpus: Brown (universal tagset)

Training sentences: 45,872 | Test sentences: 11,468

Split: 80% train / 20% test. HMM and Viterbi implemented from scratch (no library HMM/Viterbi).

Q11: Transition and Emission Probabilities (MLE + Add-1 Smoothing)

Top-5 most probable transitions from the NOUN tag:

Next Tag	P(tag NOUN)
.	0.2836
ADP	0.2439
VERB	0.1587
NOUN	0.1499
CONJ	0.0597

Top-5 most probable emissions from the VERB tag:

Word	P(word VERB)
is	0.0424
was	0.0414
be	0.0269
had	0.0214
are	0.0181

Nouns most often transition to punctuation (.) or prepositions (ADP), reflecting nominal phrases ending or being followed by PP attachments. Common verbs (is, was, be) dominate VERB emissions as auxiliary and copula forms.

Q12: Viterbi Algorithm

Implemented from scratch with a score matrix (best log-probability to each tag at each position), a backpointer matrix (best previous tag), initialization from <START>, recursion over tag transitions and emissions, termination at <END>, and backtracking to recover the optimal tag sequence.

Q13: Token-Level Accuracy

Metric	Accuracy
Overall	0.9397 (93.97%)
Seen words (in training vocab)	0.9496 (94.96%)
Unseen words (OOV)	0.4945 (49.45%)

Total test tokens: 232,177 | Seen: 227,127 | OOV: 5,050

Comment: Seen-word accuracy (94.96%) is nearly double OOV accuracy (49.45%). For seen words, the tagger has direct emission probability estimates from training. OOV

words share smoothed, uniform-like emission distributions across all tags, so the tagger relies almost entirely on transition probabilities, which is insufficient for words with ambiguous syntactic roles.

Q14: Error Analysis — 10 Mis-Tagged Tokens

Token	Gold	Predicted	Reason
average	VERB	NOUN	lexical ambiguity / sparse transition
\$200	NOUN	ADP	lexical ambiguity / sparse transition
depending	ADP	VERB	lexical ambiguity / sparse transition
very	ADV	ADJ	lexical ambiguity / sparse transition
tempting	VERB	NOUN	lexical ambiguity / sparse transition
offer	NOUN	VERB	lexical ambiguity / sparse transition
as	ADP	ADV	lexical ambiguity / sparse transition
non-partisan	ADJ	DET	OOV word
house	VERB	NOUN	lexical ambiguity / sparse transition
Spartan	NOUN	DET	lexical ambiguity / sparse transition

Error type summary: {'lexical ambiguity / sparse transition': 11437, 'OOV word': 2553}

Most frequent error type: lexical ambiguity / sparse transition (11,437 errors).

Lexical ambiguity is the dominant error source — words like 'offer' (noun/verb), 'as' (preposition/adverb), and 'house' (noun/verb) can take multiple tags depending on context. A unigram HMM cannot disambiguate these without richer features or higher-order models. OOV errors occur when hyphenated or rare compound words (e.g., 'non-partisan') lack emission estimates.